OMNI CAMERA ROVER | REV C PROTOTYPE

1. Controller

File: controller.kicad_sch

2. Motors

File: motors.kicad_sch

3. Charging

File: charging.kicad_sch

4. Regulator

File: regulator.kicad_sch

5. Interlock

File: interlock.kicad_sch

Battery protection

File: battery_protection.kicad_sch

PWR_FLAG GND

PWR_FLAG VMOTOR

PWR_FLAG USB_5V

PWR_FLAG U2_VINT

PWR_FLAG U3_VINT

PWR_FLAG CELL_P

Editable functional sheets. Global net names link sheets.
Prototype: electrical, thermal and mechanical qualification remains open.

Flex camera

Cell undervoltage load shutdown

File: camera.kicad_sch

File: cell_undervoltage.kicad_sch

Cell temperature charging window

File: charge_temperature.kicad_sch

Nathaniel Sanford

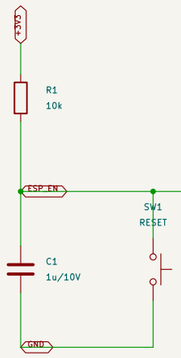
Sheet: /
File: omni_ugv.kicad_sch

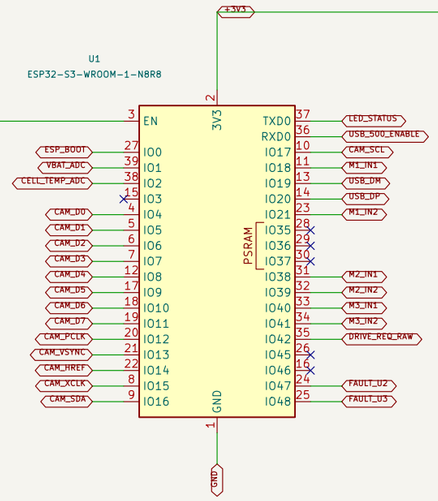
Title: Omni UGV / Rev D

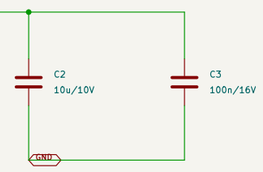
Size: A3	Date: 2026-11-06	**Rev: D – prototype**
KiCad E.D.A. 10.0.8		Id: 1/10

Controller, camera and service points

RESET / enable RC







Waveshare OV2640 | documented 2 x 9 interface

Status indicator



BOOT strap



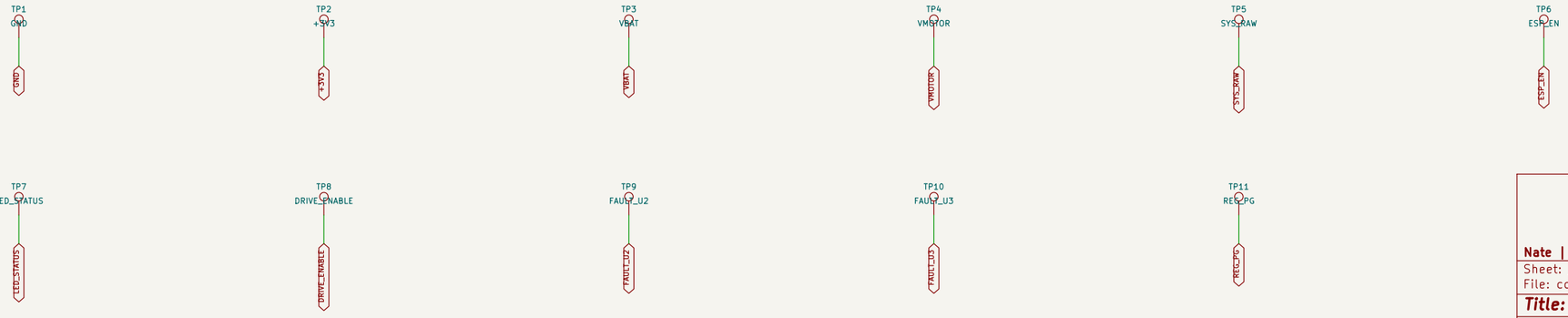
Camera reset RC





[SPEC] GPIO35 / 36 / 37 reserved for N8R8 PSRAM.

Service points (named nets continue on other sheets)



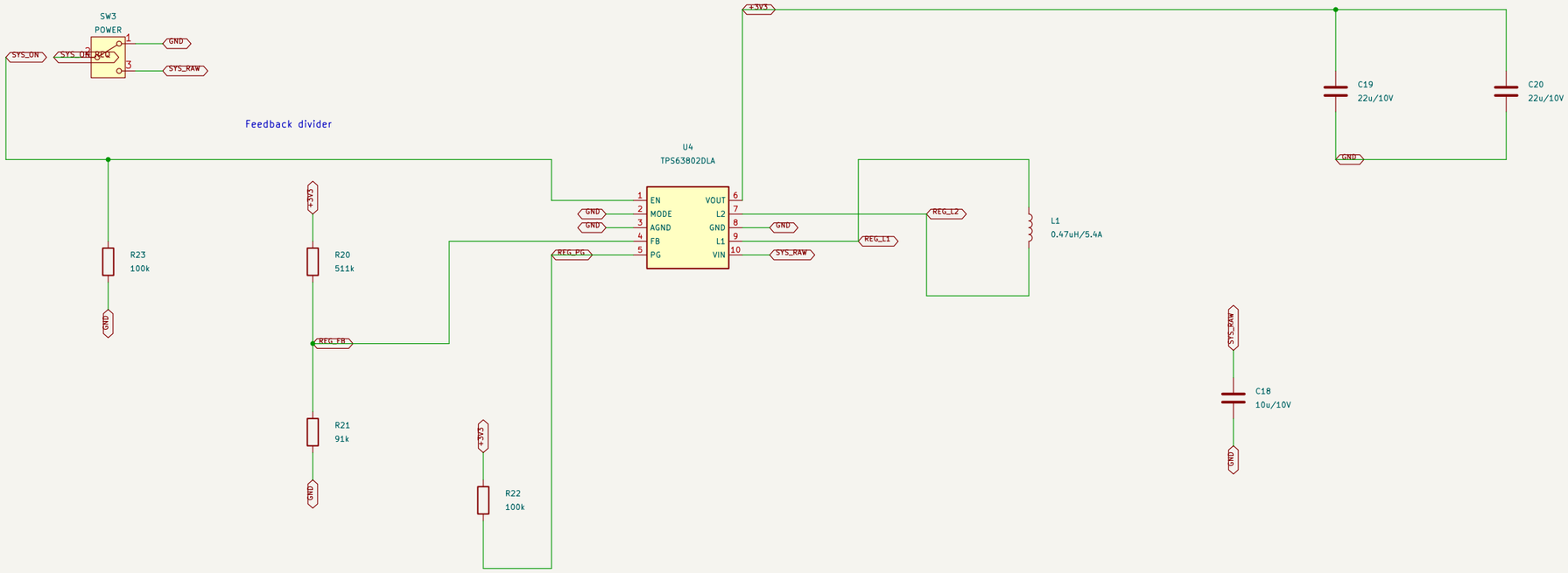
Nate | Omni camera rover	
Sheet: /1. Controller/	
File: controller.kicad_sch	
Title: Controller, camera and service points	
Size: A3	Date:
KiCad E.D.A. 10.0.8	**Rev: D — prototype**
	Id: 2/10

3.3 V buck–boost supply

System enable

Feedback divider

Output capacitors

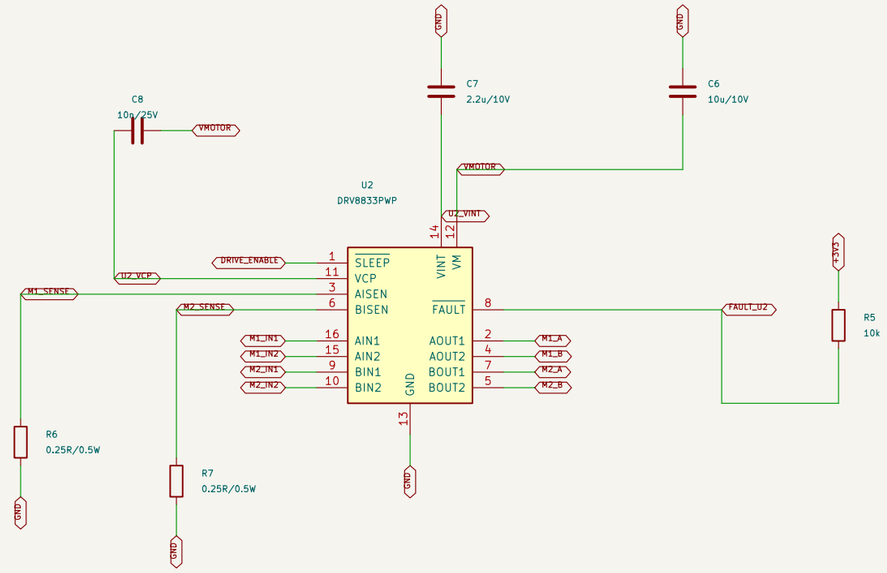


[TARGET] Keep switching loops short. Place input/output capacitors at U4.
[TBD–MEASURE] Verify startup, load transients and capacitor effective capacitance.

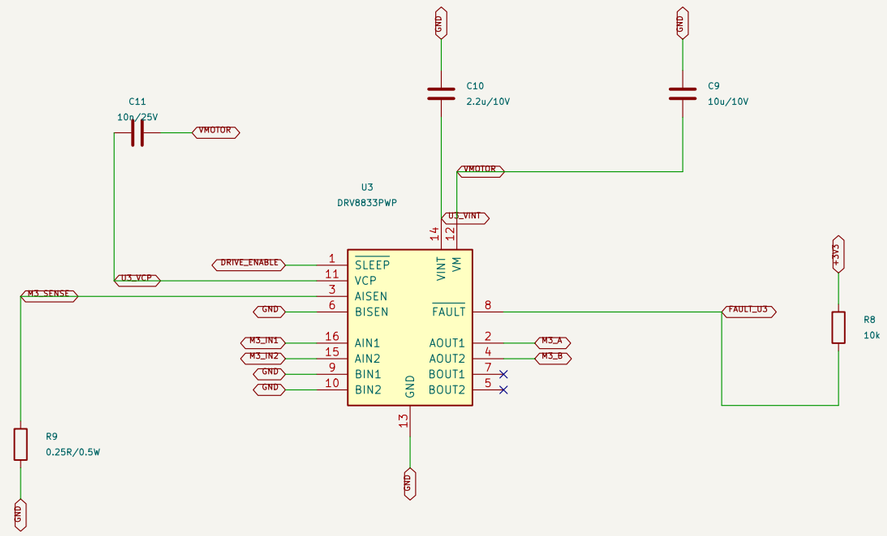
Nate | Omni camera rover		
Sheet: /4. Regulator/		
File: regulator.kicad_sch		
Title: 3.3 V buck–boost supply		
Size: A3	Date:	**Rev: D – prototype**
KiCad E.D.A. 10.0.8		Id: 3/10

Three independent motor channels

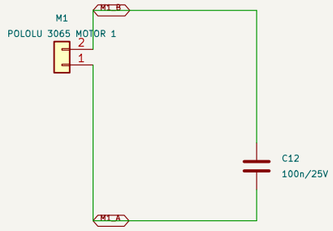
U2: wheels 1 and 2

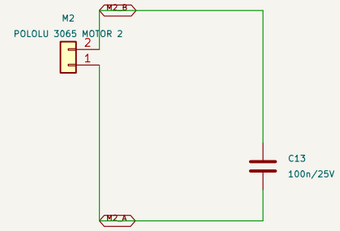


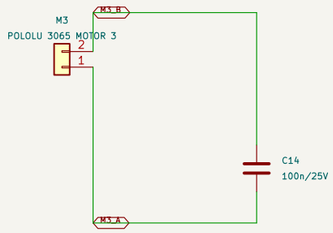
U3: wheel 3; unused bridge pins retained as designed

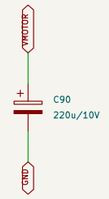


Motor terminal pads and suppression capacitors







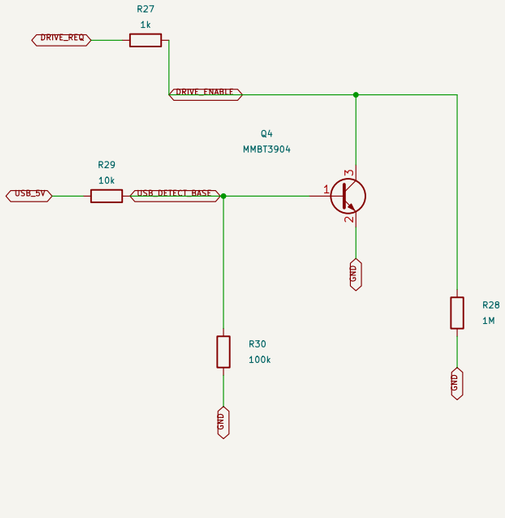


[TARGET] Route motor current away from logic returns.
[TBD-MEASURE] Braking overshoot, driver temperature and carpet load.

Nate | Omni camera rover	
Sheet: /2. Motors/	
File: motors.kicad_sch	
Title: Three independent motor channels	
Size: A3	Date:
KiCad E.D.A. 10.0.8	**Rev: D — prototype**
	Id: 4/10

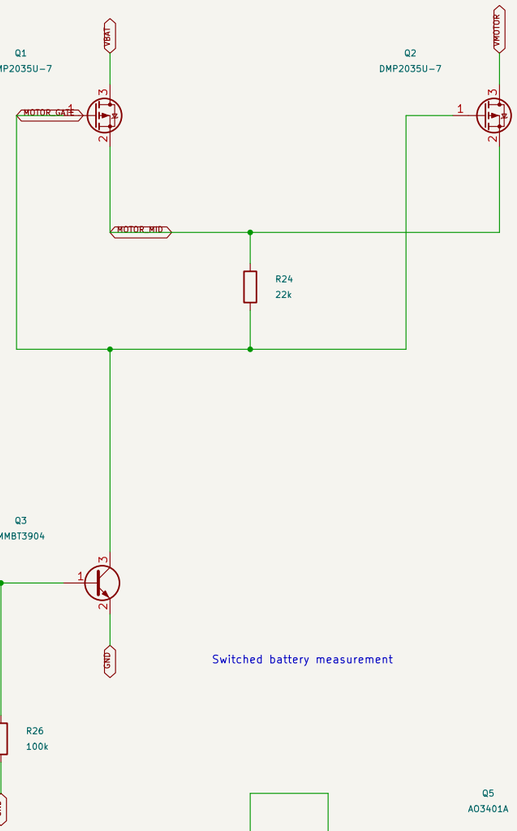
USB motor interlock and battery measurement

USB presence clamps DRIVE_ENABLE

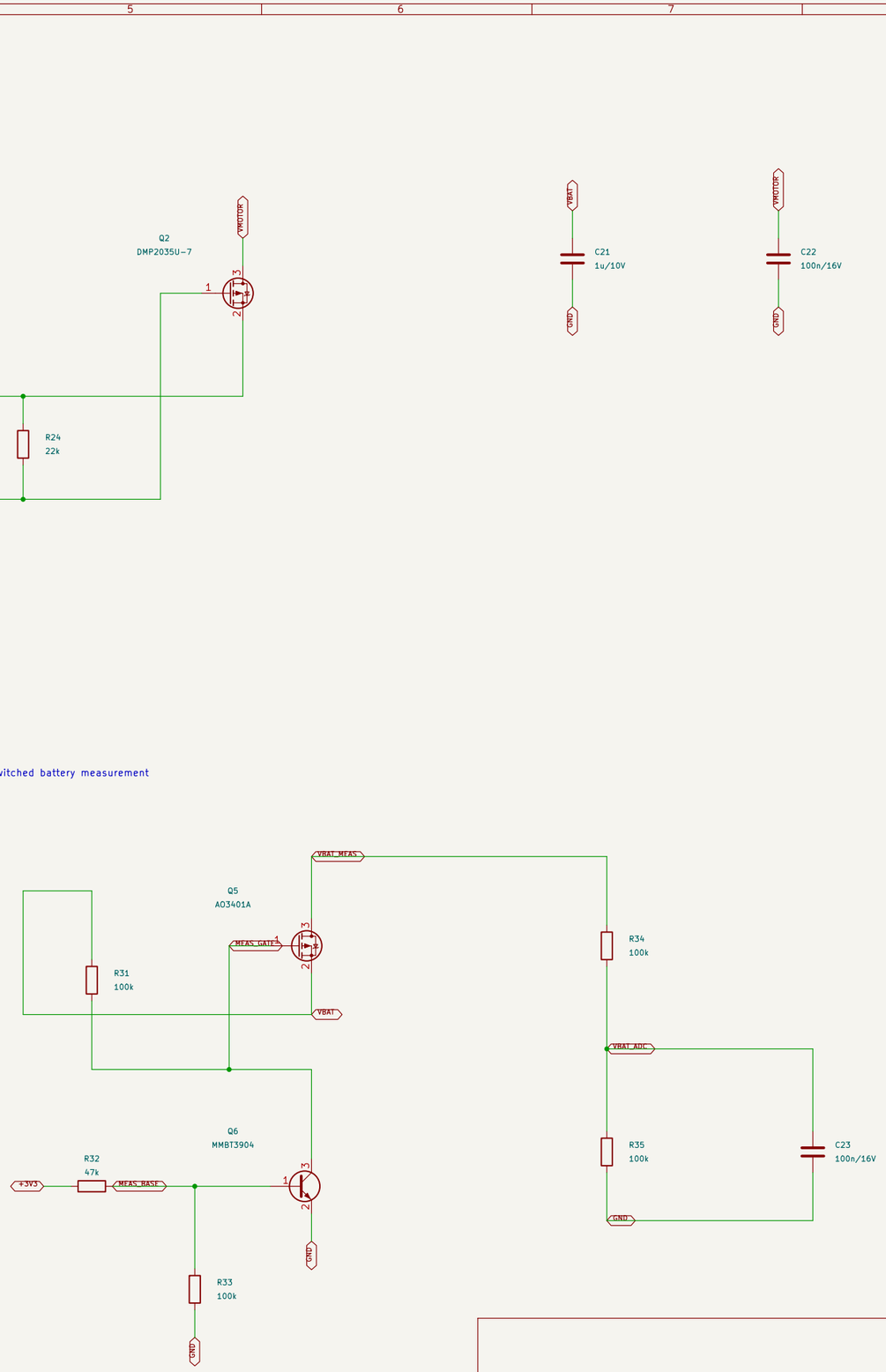


[TARGET] Reset / absent commands keep motors disabled.
[TBD-MEASURE] Verify USB insertion/removal and motor braking transients.

Back-to-back motor supply switch



Switched battery measurement



Nate | Omni camera rover	
Sheet: /5. Interlock/	
File: interlock.kicad_sch	
Title: USB motor interlock and battery measurement	
Size: A3 | Date:	**Rev: D – prototype**
KiCad E.D.A. 10.0.8	Id: 5/10

USB input, charging and cell connections

USB data to ESP32 native USB

Raw cell → protection sheet

Charge / system power path

Cell-bonded charger probe

[SPEC] J2 CELL_N is not system GND: R36 is the only intended return path.
J4: independent insulated two-wire 103AT-2 probe on the cell.
Motor current uses protected VBAT, not charger SYS_RAW output.

Nate | Omni camera rover	
Sheet: /3. Charging/	
File: charging.kicad_sch	
Title: USB input, charging and cell connections	
Size: A3	Date:
KiCad E.D.A. 10.0.8	Rev: D – prototype
	Id: 6/10

1S protection and cell-temperature monitoring

Cell voltage / current sensing

R37
100R 1%

C24
100n/16V X7R

R38
20R 1%

U7
BQ298217RUGR

R36
10m 1% 1W

R39
1k 1%

Common-drain high-side protection FET pair

R41
10M

Q8
CSD17308Q3

Q7
CSD17308Q3

R40
10M

TH2 -> J5 -> GPIO1 / ADC1

R42
10k 1%

J5
CELL NTC - DRIVE

R43
1k 1%

TH2
103AT-2 CELL-MOUNTED

C25
100n/16V X7R

TH1 -> J4 analog charge window

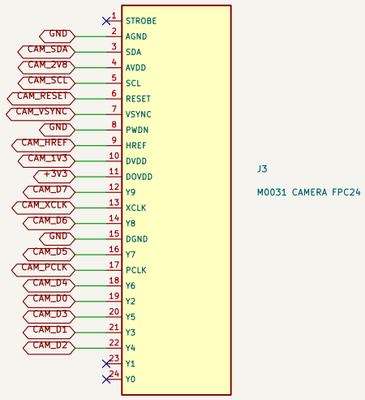
TH1
103AT-2 CELL-MOUNTED

#FLG20
PROTECTION_PWR_FLAG

#FLG21
PROTECTION_PWR_FLAG

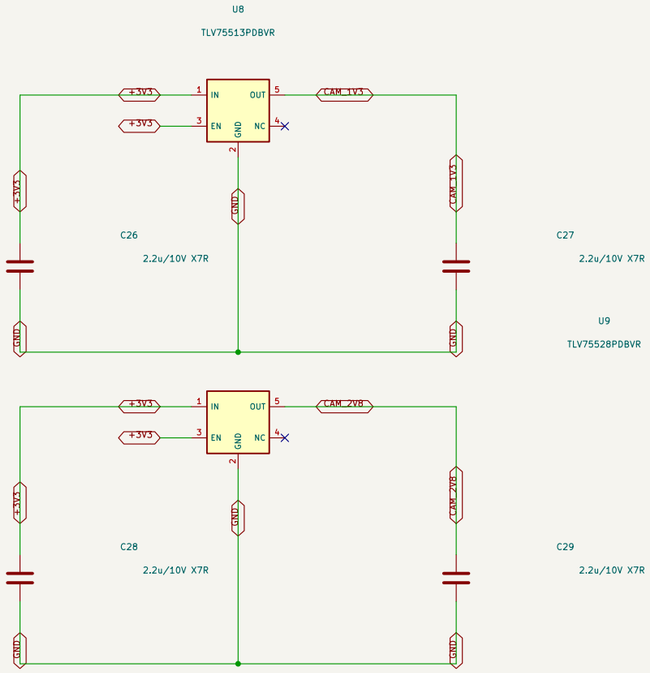
[SPEC] TH1 and TH2 are OFF-BOARD, bonded to the cell through electrical insulation.
Separate two-wire probes; do not connect either sensor to raw CELL_N.
[TARGET] Firmware must stop driving on overtemperature, open/short probe or stale readings.

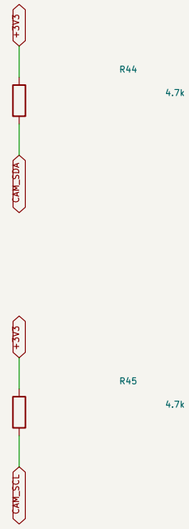
Nate | Omni camera rover	
Sheet: /Battery protection/	
File: battery_protection.kicad_sch	
Title: 1S protection and cell-temperature monitoring	
Size: A3	Date:
KiCad E.D.A. 10.0.8	Rev: D — prototype
	Id: 7/10



J3

M0031 CAMERA FPC24





M0031 only. Verify flex contact side, thickness and pin-1 before assembly.
1.3 V core / 2.8 V analog / 3.3 V I-O. No interchangeable generic OV2640 pinout.

Sheet: /Flex camera/	
File: camera.kicad_sch	
Title: OV2640 M0031 flex interface	
Size: A3	Date:
KiCad E.D.A. 10.0.8	**Rev: D — prototype**
	Id: 8/10

Independent cell undervoltage shutdown

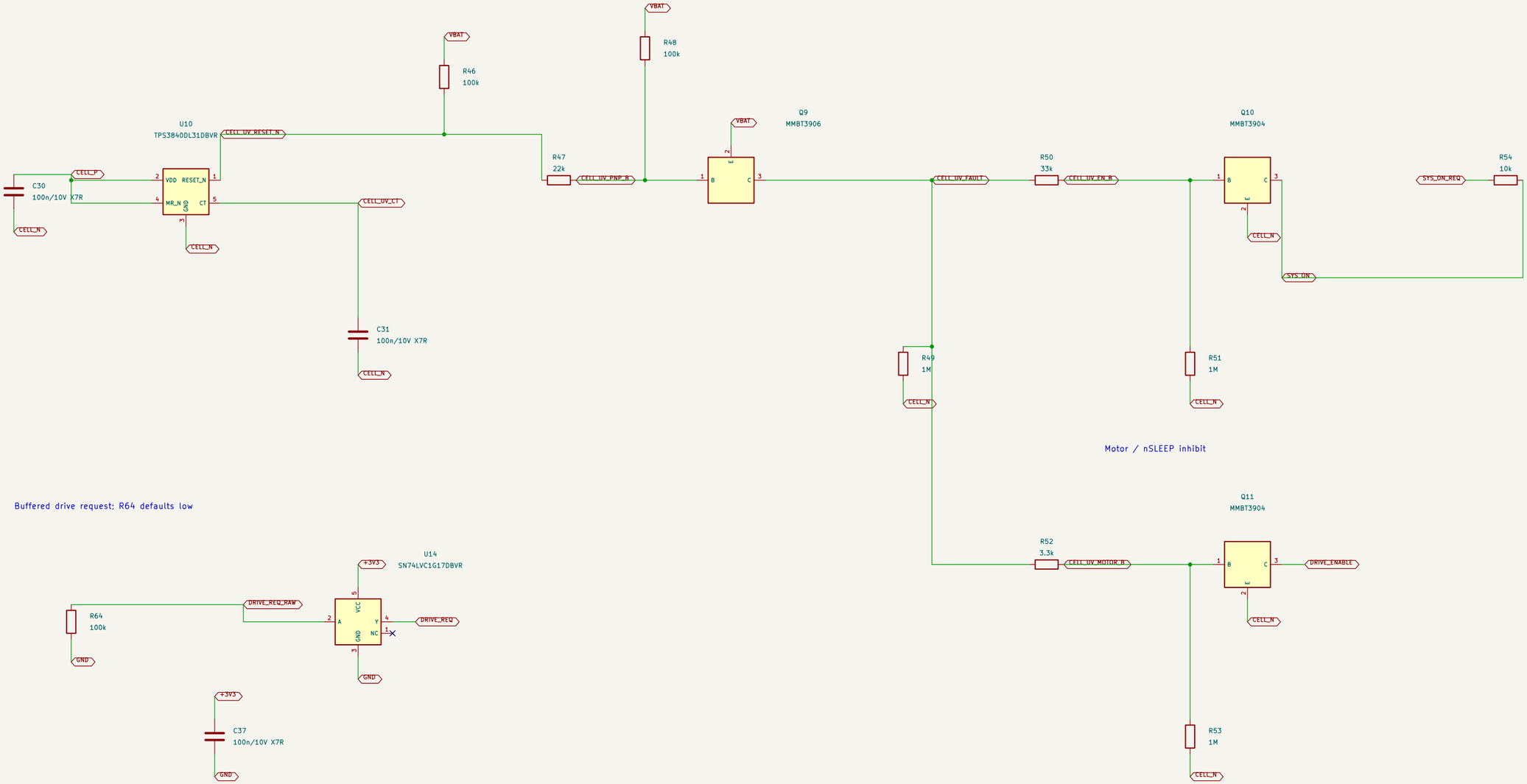
Raw-cell voltage sensing

Protected-rail inverter

Regulator enable inhibit

Motor / nSLEEP inhibit

Buffered drive request; R64 defaults low



3.1 V nominal falling cutoff; approximately 3.3 V recovery. CT delays release, not fault assertion.
Q9 and pullups use protected VBAT: only the nanopower supervisor remains on raw CELL_P after U7 cutoff.
Charging is independent of SYS_ON, so USB can recover a discharged cell while ESP32/camera stay off.
U7 retains its factory 2.6 V emergency cutoff. Firmware should stop normally before this hardware trip.

Nate | Omni camera rover		
Sheet: /Cell undervoltage load shutdown/		
File: cell_undervoltage.kicad_sch		
Title: Independent cell undervoltage shutdown		
Size: A3	Date:	**Rev: D – prototype**
KiCad E.D.A. 10.0.8		Id: 9/10

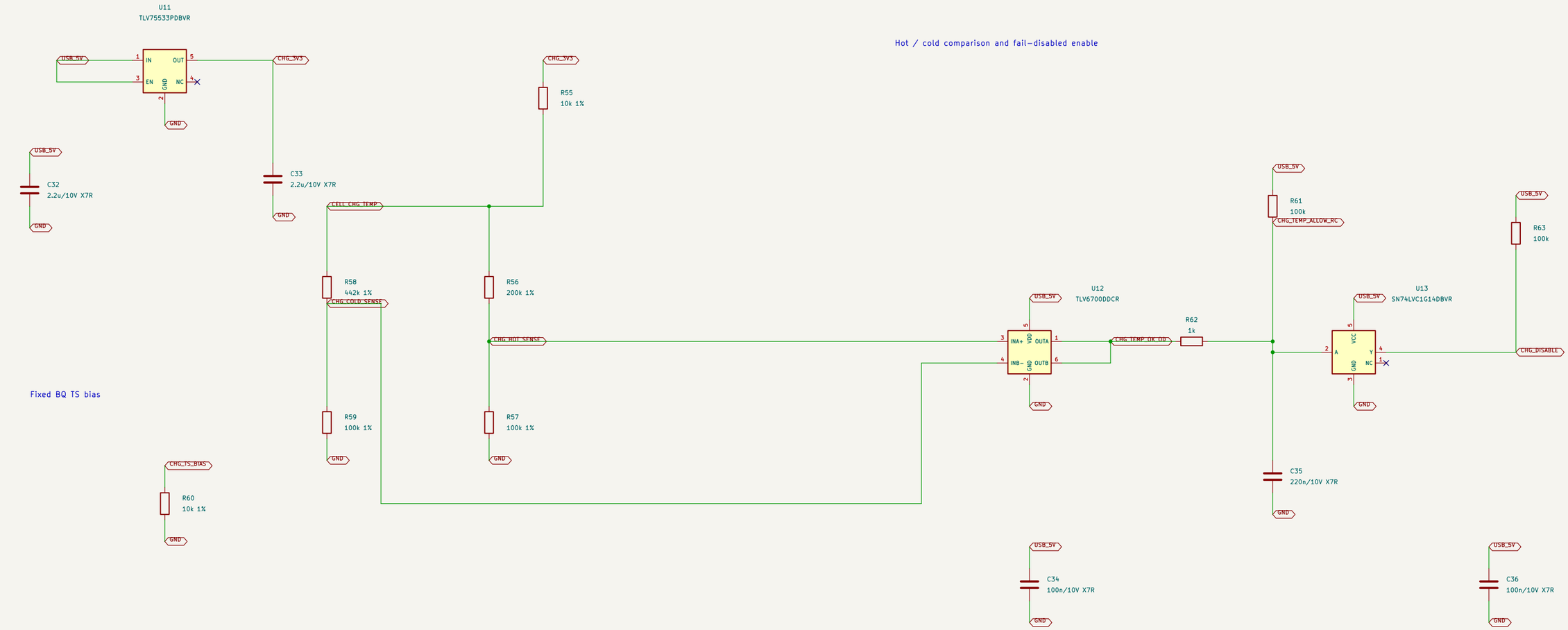
Independent cell temperature charging window

USB-powered reference supply

TH1 at J4: actual cell temperature

Hot / cold comparison and fail-disabled enable

Fixed BQ TS bias



TH1 is an electrically insulated OFF-BOARD 103AT-2, bonded to the cell. Open/short probe disables charging.
U12 open-drain outputs both release only within the temperature window. U13 then drives CE low.
R61/C35 delay charge enable at USB startup; R62 limits capacitor discharge into U12. No ESP32 dependency.
R60 intentionally replaces the direct TS thermistor. This independent circuit now decides cell charge temperature.
Tolerance calculation uses the NTC beta model: verify actual resistance-temperature curve and trip limits on a fixture.

Nate | Omni camera rover

Sheet: /Cell temperature charging window/
File: charge_temperature.kicad_sch

Title: Independent cell temperature charging window

Size: A3	Date:	**Rev: D – prototype**
KiCad E.D.A. 10.0.8		Id: 10/10